EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Let $f(x)=\begin{cases} ax+1 & \text{if } x<1 \\ 3 & \text{if } x=1 \\ bx^2+1 & \text{if } x>1 \end{cases}$. If $f(x)$ is continuous at $x = 1$ then $(a - b)$ is equal to-

(A) 0 (B) 1 (C) 2 (D) 4

MCT142

2. For the function $f(x)=\begin{cases} \dfrac{1}{x+2^{\left(\frac{1}{x-2}\right)}}, & x\neq 2 \\ k, & x=2 \end{cases}$ which of the following holds?

(A) $k = 1/2$ and f is continuous at $x = 2$ (B) $k \neq 0, 1/2$ and f is continuous at $x = 2$
(C) f can not be continuous at $x = 2$ (D) $k = 0$ and f is continuous at $x = 2$.

MCT143

3. The function $f(x)=\dfrac{4-x^2}{4x-x^3}$, is -

(A) discontinuous at only one point in its domain.
(B) discontinuous at two points in its domain.
(C) discontinuous at three points in its domain.
(D) continuous everywhere in its domain.

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MCT144

4. If $f(x)=\begin{cases} -4\sin x+\cos x & \text{for } x\le -\dfrac{\pi}{2} \\ a\sin x+b & \text{for } -\dfrac{\pi}{2}<x<\dfrac{\pi}{2} \\ \cos x+2 & \text{for } x\ge \dfrac{\pi}{2} \end{cases}$ is continuous then:

(A) $a = -1, b = 3$ (B) $a = 1, b = -3$ (C) $a = 1, b = 3$ (D) $a = -1, b = -3$

MCT145

5. The function $f(x)=\begin{cases} \dfrac{1}{4}(3x^2+1) & -\infty<x\le 1 \\ 5-4x & 1<x<4 \\ 4-x & 4\le x<\infty \end{cases}$ is -

(A) continuous at $x = 1$ & $x = 4$
(B) continuous at $x = 1$, discontinuous at $x = 4$
(C) continuous at $x = 4$, discontinuous at $x = 1$
(D) discontinuous at $x = 1$ & $x = 4$

MCT146

6. If $f(x)=\dfrac{x^2-bx+25}{x^2-7x+10}$ for $x \neq 5$ and f is continuous at $x = 5$, then $f(5)$ has the value equal to-

(A) 0 (B) 5 (C) 10 (D) 25

MCT147

For More Join: @IITJEE_Advanced

7. If $f(x)=\frac{x-e^x+\cos 2x}{x^2}, x\neq 0$ is continuous at $x = 0$, then -

(A) $f(0)=\frac{5}{2}$ (B) $[f(0)] = -2$

(C) $\{f(0)\} = -0.5$ (D) $[f(0)].\{f(0)\} = -1.5$

where [.] and {.} denotes greatest integer and fractional part function

MCT148

8. $y = f(x)$ is a continuous function such that its graph passes through $(a, 0)$.

Then $\lim\limits_{x\to a}\frac{\log_e(1+3f(x))}{2f(x)}$ is-

(A) 1 (B) 0 (C) $\frac{3}{2}$ (D) $\frac{2}{3}$

MCT149

9. In [1,3], the function $[x^2 + 1]$,[.] denoting the greatest integer function, is continuous -

(A) for all x (B) for all x except at nine points

(C) for all x except at seven points (D) for all x except at eight points

MCT150

10. Number of points of discontinuity of $f(x) = [2x^3 - 5]$ in [1,2), is equal to-

(where $[x]$ denotes greatest integer less than or equal to x)

(A) 14 (B) 13 (C) 10 (D) 8

MCT151

Pankaj Academy JEE

11. If $f(x)$ is continuous and $f\left(\frac{9}{2}\right)=\frac{2}{9}$, then the value of $\lim\limits_{x\to 0} f\left(\frac{1-\cos 3x}{x^2}\right)$ is-

(A) $\frac{2}{9}$ (B) $\frac{9}{2}$ (C) 0 (D) data insufficient

MCT152

12. f is a continuous function on the real line. Given that $x^2 + (f(x) - 2)x - \sqrt{3}\cdot f(x) + 2\sqrt{3} - 3 = 0$.

Then the value of $f(\sqrt{3})$

(A) can not be determined (B) is $2(1 - \sqrt{3})$

(C) is zero (D) is $\frac{2(\sqrt{3}-2)}{\sqrt{3}}$

MCT153

13. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :

(A) all integers (B) all integers except 0 & 1

(C) all integers except 0 (D) all integers except 1

MCT154

14. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function $\forall\ x \in \mathbb{R}$ and $f(x) = 5\ \forall\ x \in$ irrational. Then the value of $f(3)$ is -

(A) 1 (B) 2 (C) 5 (D) cannot determine

MCT155

For More Join: @IITJEE_Advanced

15. The function $f(x) = [x].\cos\frac{2x-1}{2}\pi$, where [·] denotes the greatest integer function, is discontinuous at :-

(A) all x (B) all integer points

(C) no x (D) x which is not an integer

MCT156

16. Number of points of discontinuity of $f(x)=\left\{\frac{x}{2}\right\}+\left[\frac{x}{3}\right]$ in [0, 20] is -

(where [.] & {.} denote greatest integer function and fractional part function respectively)

(A) 14 (B) 15 (C) 10 (D) 11

MCT157

17. Let $f : R \to Q$ be a continuous function such that $f(2) = 7$, then-

(A) $f(x)$ is always an even function

(B) $f(x)$ is always an odd function

(C) nothing can said about $f(x)$ being even or odd

(D) $f(x)$ is a strictly increasing function

MCT158

18. Consider $f(x)=\begin{cases}\dfrac{\sin(2\pi\sec x)}{e^x-1-x} & x\neq 0 \text{ or } (2n+1)\dfrac{\pi}{2},\ n\in I\\ k & x=0\end{cases}$

Value of k for which $f(x)$ is continuous at $x = 0$, is -

(A) $\frac{\pi}{2}$ (B) π (C) 2π (D) 4π

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MCT159

19. Let $f(x) = \begin{cases}\dfrac{e^x-e^{\sin x}}{ax^3} & ; \quad x<0\\ b & ; \quad x=0\\ \dfrac{x}{\ell n(1+4x)} & ; \quad x>0\end{cases}$. If f is continuous at $x = 0$, then $(3a + 4b)$ is equal to -

(A) –3 (B) 0 (C) 3 (D) 4

MCT160

20. The number of values of x, $x \in [-2,3]$ where $f(x) = [x^2]\sin(\pi x)$ is discontinuous is (where [.] denotes greatest integer function) -

(A) 19 (B) 8 (C) 6 (D) 0

MCT161

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. If $f(x)=\begin{cases}\sin^{-1}\left(\frac{a}{\ell nx}-\frac{b}{x-1}\right) & x\neq 1\\ \frac{\pi}{3} & x=1\end{cases}$ is continuous at $x=1$, then-

(A) $a=b$ (B) $a=\sqrt{3}$ (C) $b=\sqrt{3}$ (D) $b=-\sqrt{3}$

MCT036

2. Let $f(x)=\begin{cases}\frac{\ell n(1+x)-x+kx^2}{x^3} & ;\ x>0\\ \ell & ;\ x=0\\ \frac{\tan\frac{x}{3}}{x} & ;\ x<0\end{cases}$, then-

(A) $f(x)$ is continuous at $x=0$ if $k=\frac{1}{2},\ell=\frac{1}{3}$

(B) $f(x)$ is discontinuous at $x=0$ if $\ell\neq\frac{1}{3}$

(C) $f(x)$ has irremovable discontinuity at $x=0$ if $k\neq\frac{1}{2}$

(D) $f(x)$ has removable discontinuity at $x=0$ if $k=\frac{1}{2}$, $\ell\neq\frac{1}{3}$

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MCT037

3. Which of the following function(s) is/are discontinuous at $x=0$?

(A) $f(x)=\sin\frac{\pi}{2x}$, $x\neq 0$ and $f(0)=1$ (B) $g(x)=x\sin\left(\frac{\pi}{x}\right)$, $x\neq 0$ and $g(0)=\pi$

(C) $h(x)=\frac{|x|}{x}$, $x\neq 0$ and $h(0)=1$ (D) $k(x)=\frac{1}{1+e^{\cot x}}$, $x\neq 0$ and $k(0)=0$.

MCT038

4. A function $f(x)$ is defined as $f(x)=\frac{A\sin x+\sin 2x}{x^3},(x\neq 0)$. If the function is continuous at $x=0$, then -

(A) $A=-2$ (B) $f(0)=-1$ (C) $A=1$ (D) $f(0)=1$

MCT039

5. Which of the following function(s) can't be defined at $x=0$ to make it continuous at $x=0$?

(A) $f(x)=\frac{1}{1+2^{\frac{1}{x}}}$ (B) $f(x)=\tan^{-1}\frac{1}{x}$ (C) $f(x)=\frac{e^{\frac{1}{x}}-1}{e^{\frac{1}{x}}+1}$ (D) $f(x)=\frac{1}{\ell n|x|}$

MCT040

6. Which of the following function(s) can be defined continuously at $x=0$?

(A) $f(x)=\frac{1-\sec^2 2x}{4x^2}$ (B) $g(x)=\frac{\csc x-1}{x\csc x}$ (where $\csc x=\operatorname{cosec} x$)

(C) $h(x)=\frac{\sin 5x}{x}$ (D) $l(x)=(1+2x^2)^{\frac{1}{x^2}}$

MCT041

For More Join: @IITJEE_Advanced

7. If f is defined on an interval $[a, b]$. Which of the following statement(s) is/are INCORRECT?

(A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.

(B) If f is continuous on $[a, b]$, $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.

(C) If f is continuous on $[a, b]$ and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.

(D) If f has no zeroes on $[a, b]$, then $f(a)$ and $f(b)$ have the same sign.

MCT042

8. Which of the following functions can be defined at indicated point so that resulting function is continuous -

(A) $f(x)=\dfrac{x^2-2x-8}{x+2}$ at $x=-2$ (B) $f(x)=\dfrac{x-7}{|x-7|}$ at $x=7$

(C) $f(x)=\dfrac{x^3+64}{x+4}$ at $x=-4$ (D) $f(x)=\dfrac{3-\sqrt{x}}{9-x}$ at $x=9$

MCT043

9. In which of the following cases the given equations has atleast one root in the indicated interval ?

(A) $x - \cos x = 0$ in $(0, \pi/2)$

(B) $x + \sin x = 1$ in $(0, \pi/6)$

(C) $\dfrac{a}{x-1}+\dfrac{b}{x-3} = 0, a, b > 0$ in $(1, 3)$

(D) $f(x) - g(x) = 0$ in (a, b) where f and g are continuous on $[a, b]$ and $f(a) > g(a)$ and $f(b) < g(b)$.

MCT044

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10. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x)=\dfrac{f(x)}{g(x)}$, where h is a rational function such that

(a) Domain of $h(x)$ is $\mathbb{R} - \{-1\}$

(b) $\lim\limits_{x\to\infty} h(x) = \infty$ and

(c) $\lim\limits_{x\to -1} h(x) = \dfrac{1}{2}$.

Which of the following is/are True ?

(A) $\lim\limits_{x\to 0} (3h(x) + f(x) - 2g(x)) = -\dfrac{39}{4}$ (B) $g(x) = 4(x + 1)$

(C) $\lim\limits_{x\to 0} (3h(x) + f(x) - 2g(x)) = \dfrac{39}{4}$ (D) $g(x) = 2(x + 1)$

MCT045

11. If the function $f(x)=\dfrac{3x^2+ax+a+3}{x^2+x-2}$, $(x \neq -2)$ is continuous at $x=-2$, then

(A) $a = 15$ (B) $a = -15$ (C) $f(-2) = 1$ (D) $f(-2) = -1$

MCT046

12. Let $f(x)=\begin{cases} |ax+3| & \text{if } x\le -1 \\ |3x+a| & \text{if } -1<x\le 0 \\ \dfrac{b\sin 2x}{x}-2b & \text{if } 0<x<\pi \\ \cos^2 x-3 & \text{if } x\ge \pi \end{cases}$. If $f(x)$ is continuous for all $x \in \mathbb{R}$, then

(A) $a = 0$ (B) $a = 1$ (C) $b = 1$ (D) $b = -1$

MCT047

For More Join: @IITJEE_Advanced

13. Let $f(x)=\begin{cases}\dfrac{\ell n\cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x>0\\ \dfrac{e^{\sin 4x}-1}{\ell n(1+\tan 2x)} & \text{if } x<0\end{cases}$

(A) $f(0^+)=-2$ (B) $f(0^-)=2$

(C) $f(0^+)=2$ (D) $f(0)$ can not be defined to remove discontinuity

MCT048

14. Let $f(x)=\dfrac{\sin^{-1}(1-\{x\})\ .\ \cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\ \ (1-\{x\})}$ then

(where {.} denotes the fractional part function)

(A) $\lim\limits_{x\to 0^+} f(x)=\dfrac{\pi}{2}$ (B) f is discontinuous at $x=0$

(C) $\lim\limits_{x\to 0^-} f(x)=\dfrac{\pi}{2\sqrt{2}}$ (D) f is continuous at $x=0$

MCT049

15. Let the function $f(x)=\begin{cases}ax-b & \text{for } x\le 1\\ 3x & \text{for } 1<x<2\\ bx^2-a & \text{for } x\ge 2\end{cases}$ is continuous at $x=1$ but discontinuous at $x=2$,

then

(A) $a-b=3$ (B) $a-b=6$ (C) $b\neq 3$ (D) $b=3$

MCT050

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COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 16 to 18

Let $f(x)=\lim\limits_{n\to\infty}\dfrac{\ell nx+x^n}{(x-1)+x^n}\ (x>0)$.

On the basis of above information, answer the following questions:

16. $\lim\limits_{x\to 1^-}\dfrac{\sin(f(x)-1)}{(x-1)}$ is equal to -

(A) 0 (B) $\dfrac{1}{2}$ (C) $-\dfrac{1}{2}$ (D) 1

MCT053

17. For $f(x)$ which of the following statement is true -

(A) $f(x)$ is discontinuous at $x=1$

(B) non-negative difference between LHL & RHL at $x=1$ is equal to 2.

(C) $f(x)$ is many-one function

(D) $f(x)$ is one-one function

MCT054

18. The number of solutions of the equation $f(x)=\dfrac{1}{(x-1)^2}$ is -

(A) 0 (B) 1 (C) 2 (D) 3

MCT055

For More Join: @IITJEE_Advanced

Paragraph for Question No. 19 to 20

Consider $f(x)=\frac{1}{\ell n|x|}$ and $g(x) = \tan^{-1}|x|$.

On the basis of above information, answer the following questions :

19. Sum of solutions of the equation $\frac{g(x)}{f(x)}=\frac{1}{2}$ will be less than or equal to :

(A) 0 (B) 1 (C) 2 (D) 3

MCT062

20. The value(s) of x at which $h(x) = f(x) + g(x)$ will be discontinuous is/are :

(A) 0 (B) 1 (C) –1 (D) 2

MCT063

MATRIX MATCH TYPE QUESTION

21.

	Column-I		**Column-II**
(A)	$\lim_{x\to1}\frac{x^3-1}{\ell nx}$ is	(P)	2
(B)	$\lim_{x\to0}\frac{x(\cos x-\cos 2x)}{2\sin x-\sin 2x}$ is	(Q)	3
(C)	$\lim_{x\to0}\frac{\tan x\sqrt{\tan x}-\sin x\sqrt{\sin x}}{x^3.\sqrt{x}}$ is	(R)	$\frac{3}{2}$
(D)	If $f(x) = \cos(x\cos\frac{1}{x})$ and $g(x) =\frac{\ell n(\sec^2 x)}{x\sin x}$ are both continuous at $x = 0$ then $f(0) + g(0)$ equals	(S)	$\frac{3}{4}$

MCT064

MATCHING LIST TYPE QUESTION

22. Match the function in column-I with its behaviour at $x = 0$ in column-II, where [.] denotes greatest integer function & $sgn(x)$ denotes signum function.

	Column-I		**Column-II**
(A)	$f(x) = [x][1 + x]$	(P)	LHL exist at $x = 0$
(B)	$f(x) = [-x][1 + x]$	(Q)	RHL exist at $x = 0$
(C)	$f(x) = (\text{sgn}(x))[2 - x][1 + |x|]$	(R)	Continuous at $x = 0$
(D)	$f(x) = [\cos x]$	(S)	$\lim_{x\to0} f(x)$ exists but function is discontinuous at $x = 0$
		(T)	$\lim_{x\to0} f(x)$ does not exist

MCT065

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For More Join: @IITJEE_Advanced

EXERCISE (S)

1. The function $f(x)=\begin{cases}\left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if} \quad 0<x<\frac{\pi}{2}\\ b+2 & \text{if} \quad x=\frac{\pi}{2}\\ (1+|\cos x|)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if} \quad \frac{\pi}{2}<x<\pi\end{cases}$

If $f(x)$ is continuous at $x=\pi/2$, then the value of $(a-b)$ is

MCT066

2. Let $f(x)=\begin{cases}\frac{1-\sin \pi x}{1+\cos 2\pi x}, & x<\frac{1}{2}\\ p, & x=\frac{1}{2}\\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x>\frac{1}{2}\end{cases}$. The number of all possible values of 'p' so that the function $f(x)$ is continuous at $x=1/2$, is –

MCT067

3. Let $f(x)=\begin{bmatrix}1+x, & 0\le x\le 2\\ 3-x, & 2<x\le 3\end{bmatrix}$ and $g(x)=f[f(x)]$ & hence the number of points of discontinuity of $g(x)$ is

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MCT068

4. If $f(x)$ is continuous at $x=\frac{\pi}{2}$ where $f(x)=\begin{cases}\frac{1-\sin^3 x}{3\cos^2 x} & \text{if} \quad x<\frac{\pi}{2}\\ a & \text{if} \quad x=\frac{\pi}{2},\\ \frac{b(1-\sin x)}{(\pi-2x)^2} & \text{if} \quad x>\frac{\pi}{2}\end{cases}$

then the value of $(4a+3b)$ is

MCT069

5. If the function $f(x)=\begin{cases}\frac{\sin(a+1)x+\sin x}{x} & \text{for} \quad x<0\\ c & \text{for} \quad x=0\\ \frac{(x+bx^2)^{1/2}-x^{1/2}}{bx^{3/2}} & \text{for} \quad x>0\end{cases}$ is continuous at $x=0$, then the value of $(2a+14c)$ is

MCT070

6. If $f(x)=\frac{\sin 3x+A\sin 2x+B\sin x}{x^5}$ $(x\neq 0)$ is continuous at $x=0$, then the value of $(A+B+f(0))$ is

MCT071

For More Join: @IITJEE_Advanced

7. If $f(x) = x + \{-x\} + [x]$, where $[x]$ is the integral part & $\{x\}$ is the fractional part of x, then the number of points of discontinuity of $f(x)$ in $[-2, 2]$ is

MCT072

8. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\cos ecx} & ; \quad \frac{-\pi}{2} < x < 0 \\ a & ; \quad x = 0 \\ \frac{e^{1/x} + e^{2/x} + e^{3/|x|}}{ae^{2/x} + be^{3/|x|}} & ; \quad 0 < x < \frac{\pi}{2} \end{cases}$

If $f(x)$ is continuous at $x = 0$, then the value of $(a^2 \cdot b^2)$ is

MCT073

9. Let f be a real valued continuous function on $\mathbb{R}$ and satisfying $f(-x) - f(x) = 0 \ \forall \in \mathbb{R}$. If $f(-5) = 5$, $f(-2) = 4$, $f(3) = -2$ and $f(0) = 0$ then find the minimum number of zero's of the equation $f(x) = 0$.

MCT074

10. Find the number of points of discontinuity of the function $f(x) = [5x] + \{3x\}$ in $[0, 5]$ where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively.

MCT075

11. If $f(x) = \begin{cases} \frac{1 - \sin^3 x}{3\cos^2 x} & , \ x < \pi/2 \\ a & , \ x = \pi/2 \\ \frac{b(1 - \sin x)}{(\pi - 2x)^2} & , \ x > \pi/2 \end{cases}$ is continuous at $x = \frac{\pi}{2}$. Then the value of $2a + b$ is

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MCT095

12. If $g(x) = \left(|x-1| + |4x-11|\right)\left[x^2 - 2x - 2\right]$, then find the number of point of discontinuity of $g(x)$ in $\left(\frac{1}{2}, \frac{5}{2}\right)$

{where [.] denotes GIF}

MCT096

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let $f: R \to R$ be a function defined as : $f(x) = \begin{cases} 5, & \text{if } x \le 1 \\ a+bx, & \text{if } 1 < x < 3 \\ b+5x, & \text{if } 3 \le x < 5 \\ 30, & \text{if } x \ge 5 \end{cases}$. Then, f is :

[JEE (Main) 2019]

(1) continuous if $a = 5$ and $b = 5$ (2) continuous if $a = -5$ and $b = 10$

(3) continuous if $a = 0$ and $b = 5$ (4) not continuous for any values of a and b

MCT025

2. If the function f defined on $\left(\frac{\pi}{6}, \frac{\pi}{3}\right)$ by $f(x) = \begin{cases} \frac{\sqrt{2}\cos x - 1}{\cot x - 1}, & x \ne \frac{\pi}{4} \\ k, & x = \frac{\pi}{4} \end{cases}$ is continuous, then k is equal to

[JEE (Main) 2019]

(1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{\sqrt{2}}$ (4) 2

MCT026

3. If the function $f(x) = \begin{cases} a|\pi - x| + 1, & x \le 5 \\ b|x - \pi| + 3, & x > 5 \end{cases}$ is continuous at $x = 5$, then the value of $a - b$ is :-

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[JEE (Main) 2019]

(1) $\frac{2}{5-\pi}$ (2) $\frac{2}{\pi-5}$ (3) $\frac{2}{\pi+5}$ (4) $\frac{-2}{\pi+5}$

MCT027

4. Let [t] denote the greatest integer $\le$ t and $\lim\limits_{x \to 0} x\left[\frac{4}{x}\right] = A$. Then the function, $f(x) = [x^2]\sin(\pi x)$ is discontinuous, when x is equal to :

[JEE (Main) 2020]

(1) $\sqrt{A+5}$ (2) $\sqrt{A+1}$ (3) $\sqrt{A}$ (4) $\sqrt{A+21}$

MCT028

5. Let $f(x) = x \cdot \left[\frac{x}{2}\right]$, for $-10 < x < 10$, where $[t]$ denotes the greatest integer function. Then the number of points of discontinuity of f is equal to.

[JEE (Main) 2020]

MCT029

6. Let $f: R \to R$ and $g: R \to R$ be defined as

$f(x) = \begin{cases} x + a, & x < 0 \\ |x - 1|, & x \ge 0 \end{cases}$ and $g(x) = \begin{cases} x + 1, & x < 0 \\ (x-1)^2 + b, & x \ge 0 \end{cases}$

where a, b are non-negative real numbers. If $(gof)(x)$ is continuous for all $x \in R$, then $a + b$ is equal to.

[JEE (Main) 2021]

MCT030

For More Join: @IITJEE_Advanced

7. If the function $f(x)=\frac{\cos(\sin x)-\cos x}{x^4}$ is continuous at each point in its domain and $f(0)=\frac{1}{k}$, then k is. **[JEE (Main) 2021]**

MCT031

8. Let a function $f: \boldsymbol{R} \to \boldsymbol{R}$ be defined as

$$f(x)=\begin{cases}\sin x-e^x & \text{if } x \le 0\\ a+[-x] & \text{if } 0<x<1\\ 2x-b & \text{if } x \ge 1\end{cases}$$

Where $[x]$ is the greatest integer less than or equal to x. If f is continuous on **R**, then $(a+b)$ is equal to: **[JEE (Main) 2021]**

(1) 4 (2) 3 (3) 2 (4) $5d$

MCT032

9. The number of points where the function

$$f(x)=\begin{cases}|2x^2-3x-7| & \text{if } x \le -1\\ [4x^2-1] & \text{if } -1<x<1\\ |x+1|+|x-2| & \text{if } x \ge 1\end{cases}$$

$[t]$ denotes the greatest integer $\le t$, is discontinuous is. **[JEE (Main) 2022]**

MCT033

10. Let $f(x)=[2x^2+1]$ and $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x \ge 0\end{cases}$, where $[t]$ is the greatest integer $\le t$. Then, in the open interval $(-1, 1)$, the number of points where fog is discontinuous is equal to. **[JEE (Main) 2022]**

Rankers Academy JEE

MCT034

11. Let $f: \mathbb{R} \to \mathbb{R}$ be defined as

$$f(x)=\begin{cases}[e^x], & x<0\\ ae^x+[x-1], & 0 \le x<1\\ b+[\sin(\pi x)], & 1 \le x<2\\ [e^{-x}]-c, & x \ge 2\end{cases}$$

where $a, b, c \in \mathbb{R}$ and $[t]$ denotes greatest integer less than or equal to t. Then, which of the following statements is true ? **[JEE (Main) 2022]**

(1) There exists $a, b, c \in \mathbb{R}$ such that f is continuous of $\mathbb{R}$.

(2) If f is discontinuous at exactly one point, then $a+b+c=1$.

(3) If f is discontinuous at exactly one point, then $a+b+c \ne 1$.

(4) f is discontinuous at atleast two points, for any values of a, b and c.

MCT035

12. If the function $f(x)=\begin{cases}\frac{\log_e(1-x+x^2)+\log_e(1+x+x^2)}{\sec x-\cos x}, & x \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)-\{0\}\\ k, & x=0\end{cases}$

is continuous at $x = 0$, then k is equal to : **[JEE (Main) 2022]**

(1) 1 (2) –1 (3) e (4) 0

MCT076

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13. Let $f, g: \mathbf{R} \to \mathbf{R}$ be functions defined by

$$f(x)=\begin{cases}[x] & , \quad x<0 \\ |1-x| & , \quad x \geq 0\end{cases} \text{ and}$$

$$g(x)=\begin{cases}e^x-x & , \quad x<0 \\ (x-1)^2-1 & , \quad x \geq 0\end{cases}$$

where $[x]$ denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly : **[JEE (Main) 2022]**

(1) one point (2) two points (3) three points (4) four points

MCT077

14. Let $f(x)=[x^2-x]+|-x+[x]|$, where $x \in \mathbb{R}$ and $[t]$ denotes the greatest integer less than or equal to t. Then, f is **[JEE (Main) 2023]**

(1) continuous at $x = 0$, but not continuous at $x = 1$

(2) continuous at $x = 0$ and $x = 1$

(3) not continuous at $x = 0$ and $x = 1$

(4) continuous at $x = 1$, but not continuous at $x = 0$

MCT060

15. Let $[x]$ be the greatest integer $\leq x$. Then the number of points in the interval (–2,1), where the function $f(x) = |[x]| + \sqrt{x-[x]}$ is discontinuous is. **[JEE (Main) 2023]**

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MCT061

16. If the function $f(x)=\begin{cases}\dfrac{72^x-9^x-8^x+1}{\sqrt{2}-\sqrt{1+\cos x}} & , \quad x \neq 0 \\ a\log_e 2\log_e 3 & , \quad x=0\end{cases}$ is continuous at $x = 0$, then the value of a^2 is equal to

[JEE (Main) 2024]

(1) 968 (2) 1152 (3) 746 (4) 1250

MCT114

17. Let $f: [-1, 2] \to R$ be given by $f(x) = 2x^2 + x + [x^2] - [x]$, where $[t]$ denotes the greatest integer less than or equal to t. The number of points, where f is not continuous, is : **[JEE (Main) 2024]**

(1) 6 (2) 3 (3) 4 (4) 5

MCT115

18. Let $[t]$ denote the greatest integer less than or equal to t. Let $f: [0, \infty) \to R$ be a function defined by $f(x)=\left[\frac{x}{2}+3\right]-[\sqrt{x}]$. Let S be the set of all points in the interval $[0, 8]$ at which f is not continuous. Then $\sum_{a \in S} a$ is equal to _____. **[JEE (Main) 2024]**

MCT116

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19. Lef $f:(0,\pi) \to R$ be a function given by

$$f(x)=\begin{cases} \left(\frac{8}{7}\right)^{\frac{\tan 8x}{\tan 7x}}, & 0<x<\frac{\pi}{2} \\ a-8, & x=\frac{\pi}{2} \\ (1+|\cot x|)^{\frac{b}{a}|\tan x|}, & \frac{\pi}{2}<x<\pi \end{cases}$$

Where $a, b \in Z$. If f is continuous at $x=\frac{\pi}{2}$, then $a^2 + b^2$ is equal to _____. **[JEE (Main) 2024]**

MCT117

20. Consider the function.

$$f(x)=\begin{cases} \frac{a\left(7x-12-x^2\right)}{b\left|x^2-7x+12\right|}, & x<3 \\ 2^{\frac{\sin(x-3)}{x-[x]}}, & x>3 \\ b, & x=3 \end{cases}$$

Where $[x]$ denotes the greatest integer less than or equal to x. If S denotes the set of all ordered pairs (a, b) such that $f(x)$ is continuous at $x = 3$, then the number of elements in S is :

[JEE (Main) 2024]

(1) 2 (2) Infinitely many (3) 4 (4) 1

MCT118

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EXERCISE - JEE (Advanced) PYQ

1. For every integer n, let a_n and b_n be real numbers. Let function $f:\mathbb{R} \to \mathbb{R}$ be given by

$$f(x)=\begin{cases} a_n+\sin \pi x, & \text{for} \quad x\in[2n,2n+1] \\ b_n+\cos \pi x, & \text{for} \quad x\in(2n-1,2n) \end{cases}$$, for all integers n.

If f is continuous, then which of the following holds(s) for all n? **[IIT JEE 2012]**

(A) $a_{n-1}-b_{n-1}=0$ (B) $a_n-b_n=1$ (C) $a_n-b_{n+1}=1$ (D) $a_{n-1}-b_n=-1$

MCT057

2. For every pair of continuous function $f,g:[0,1]\to\mathbb{R}$ such that

$\max\{f(x):x\in[0,1]\}=\max\{g(x):x\in[0,1]\}$,

the correct statement(s) is(are) : **[JEE (Advanced) 2014]**

(A) $(f(c))^2+3f(c)=(g(c))^2+3g(c)$ for some $c\in[0,1]$

(B) $(f(c))^2+f(c)=(g(c))^2+3g(c)$ for some $c\in[0,1]$

(C) $(f(c))^2+3f(c)=(g(c))^2+g(c)$ for some $c\in[0,1]$

(D) $(f(c))^2=(g(c))^2$ for some $c\in[0,1]$

MCT058

3. Let $[x]$ be the greatest integer less than or equal to x. Then, at which of the following point(s) the function $f(x)=x\cos(\pi(x+[x]))$ is discontinuous ? **[JEE (Advanced) 2017]**

(A) $x=-1$ (B) $x=0$ (C) $x=2$ (D) $x=1$

MCT059

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	C	D	A	B	A	D	C	D	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	C	C	C	A	C	C	B

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C	A,B,C,D	A,B,C,D	A,B	A,B,C	A,C,D	A,C,D	A,C,D	A,B,C,D	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,D	A,C	A,B,D	A,B,C	A,C	C	C	C	A,B,C,D	A,B,C
Que.	21					22				
Ans.	(A→Q; B→R; C→S; D→P)					(A→P,Q,R; B→P,Q,T; C→P,Q,T; D→P,Q,S)				

EXERCISE (S)

1. 1 **2.** 0 **3.** 2 **4.** 14 **5.** 4
6. 2.00 **7.** 5.00 **8.** 1.00 **9.** 5.00 **10.** 30.00
11. 5 **12.** 2

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EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	1	1	2	8	1	6	2	7	62
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	1	2	4	2	2	3	17	81	4

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	
Ans.	B,D	A,D	A,C,D	

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